

TEDDY MBAYAKI

Software Engineer

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PROFESSIONAL PROFILE

Software Engineer experienced in designing, developing, and maintaining scalable backend systems and APIs. Built and shipped production frontend and backend applications with real LLM integrations through the FlyRank AI internship — using React, Next.js (App Router), and TypeScript on the frontend, and Node.js, Express, PostgreSQL, Docker, and Stripe on the backend to ship features, debug live deployment issues, run accessibility and performance audits, and write tests. Proficient in building cloud-native applications, managing SQL and NoSQL databases, and deploying applications on Vercel. Strong problem-solving and troubleshooting abilities, with a proven track record of collaborating with cross-functional teams to deliver reliable, high-quality software solutions.

PROFESSIONAL EXPERIENCE

Frontend AI Engineering, Backend AI Engineering & AI Fluency Intern — FlyRank AI Internship

Jun 2026 – 30 Aug 2026 | Remote

Frontend AI Engineering

- Migrated a weather dashboard capstone project from React/Vite to React, Next.js (App Router), and TypeScript, scaffolding routed pages and deploying to Vercel with automatic preview builds on every push.
- Built a streaming, multi-turn AI chat feature (Vercel AI SDK + Google Gemini) grounded in live weather data, including a thinking-state indicator, a working stop/resume flow, and mobile-responsive layout built with React and Tailwind CSS.
- Diagnosed and resolved a production deployment failure where the build succeeded and every route was listed, but the live site 404'd — traced to a misconfigured Vercel Framework Preset setting, not application code.
- Improved Lighthouse mobile performance score from 49 to 85 (a 73% increase) by lazy-loading a non-critical AI SDK bundle, and resolved all WCAG 2.1 AA accessibility violations from 2 down to zero, verified with axe DevTools.
- Wrote 4 passing unit tests (Vitest, React Testing Library) covering 2 core UI components as part of a test-driven verification pass before deployment.
- Practiced and documented structured prompt-engineering techniques — role assignment, context framing, few-shot examples, output structuring, step decomposition — across real debugging and content-generation tasks, with cross-model comparison (Claude vs. ChatGPT).

Backend AI Engineering

- Built a REST CRUD API (Node.js, Express) with Zod-style input validation, correct status codes (200/201/204/400/404), and Swagger/OpenAPI documentation, then migrated its storage layer across three engines — in-memory → SQLite → containerized PostgreSQL — keeping endpoints and behavior identical and using parameterized queries throughout.
- Containerized the API and PostgreSQL with Docker and Docker Compose, using named volumes for persistence across full teardown/restart and a database healthcheck (depends_on: service_healthy) to fix a startup race condition; kept secrets in git-ignored .env files with a committed .env.example.
- Implemented authentication with Supabase Auth — signup, login, logout — and JWT verification through reusable Express middleware protecting multiple routes, returning 401 for missing, tampered, or expired tokens; documented bearer auth with an Authorize flow in Swagger UI.
- Built a polite web-scraping pipeline (fetch → extract → normalize → validate → store → report) in Node.js with Cheerio and Zod: honest user-agent, timeouts, request delays, HTML caching, relative-to-absolute URL resolution, schema-validated output with a rejected-records file, idempotent reruns (exactly 60 unique records across 3 catalogue pages), single-retry rules for 5xx/timeouts (never 404/403), and a per-run report that stays accurate when a page fails.
- **Capstone — Usage Metering & Billing Engine:** designed exactly-once usage metering using a database-level unique constraint on (tenant, idempotency key); enforced plan quotas with exact boundary behavior and 429 responses; calculated costs in integer cents with AI-token pricing rules (cheaper cached input, reasoning tokens billed as output); integrated Stripe test-mode Checkout with signature-verified, deduplicated webhooks that sync tenant plans; added a background usage-alert job with retries.

- **Capstone — Embeddable Widget & Lead-Capture Platform:** built a multi-tenant widget management API with proven tenant isolation (cross-tenant access returns 404); a public cross-origin submission endpoint with CORS preflight handling, boundary validation, per-IP rate limiting, honeypot spam control, a geolocation provider fallback chain (provider A → B → none) proven deterministically with mocks, and a failure-tolerant notification side effect; cached, versioned embeddable widget script; and an owner dashboard API with per-widget and geo analytics.
- Documented every requirement with request/response evidence and an honest AI-usage build log; diagnosed and fixed real bugs along the way, including a cache-key collision, a CORS preflight failure, and an accidentally committed `.env` (repository history cleaned and verified).

Customer Service Assistant — Cyber Café / Internet Café

Jan 2026 – Apr 2026 | Kenya

- Provided customer support for printing, scanning, photocopying, and internet access services.
- Handled document processing, including typing, formatting, editing, and printing of letters, CVs, and reports.
- Managed data entry and record keeping using Microsoft Word and Excel.
- Assisted customers with file uploads, downloads, and online form submissions while safeguarding customer data.

IT & Administrative Support Intern — The Eldoret National Polytechnic

May 2024 – Jul 2024 | Kenya

- Assisted with network setup, configuration, and troubleshooting across campus systems.
- Supported network security by managing user access and monitoring connectivity.
- Helped debug software issues and resolve system errors for university applications.
- Improved network uptime by 15% through faster troubleshooting and performance monitoring.

TECHNICAL SKILLS

Frontend & AI: React, Next.js (App Router), TypeScript, Tailwind CSS, Vercel AI SDK, LLM integration (Claude, Gemini) — used in the FlyRank internship above

Backend Development: Node.js, Express, Python, Java, Microservices Architecture; REST API design, JWT authentication (Supabase Auth), Stripe (test mode) payments and webhooks, idempotency, rate limiting, CORS, web scraping (Cheerio), schema validation (Zod) — backend items used in the FlyRank internship above

API Development: RESTful APIs, GraphQL, Swagger/OpenAPI

Databases: PostgreSQL, SQLite, MySQL, MongoDB, DynamoDB, SQL Design & Optimization

Cloud Platforms: Vercel

DevOps & Containers: Docker, Docker Compose, Kubernetes

Testing & Quality: Vitest, React Testing Library, Lighthouse, axe DevTools (WCAG 2.1 AA) — used in the FlyRank internship above

Operating Systems: Linux System Administration

Version Control: Git & GitHub

EDUCATION

Bachelor of Science in Computer Science

Catholic University of Eastern Africa, Nairobi, Kenya — Sep 2022 – Oct 2026

LANGUAGES

English — Fluent (advanced communication & technical writing) | Kiswahili — Fluent (native speaker)